

The power of offloading.

What the chat logs show, and
what I changed in my own course.

EEG during essay writing

MIT Media Lab, 2025 · 54 students

Brain only strongest connectivity

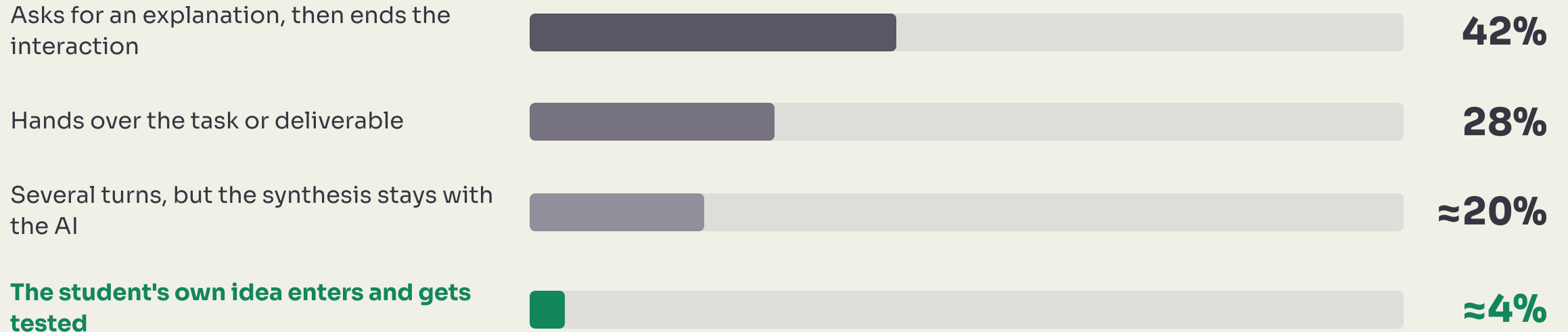
Search engine in between

LLM **weakest**

The LLM group showed weaker brain connectivity.

In search of constructive use

40,000 student messages to an AI tutor



Preliminary, descriptive categorisation

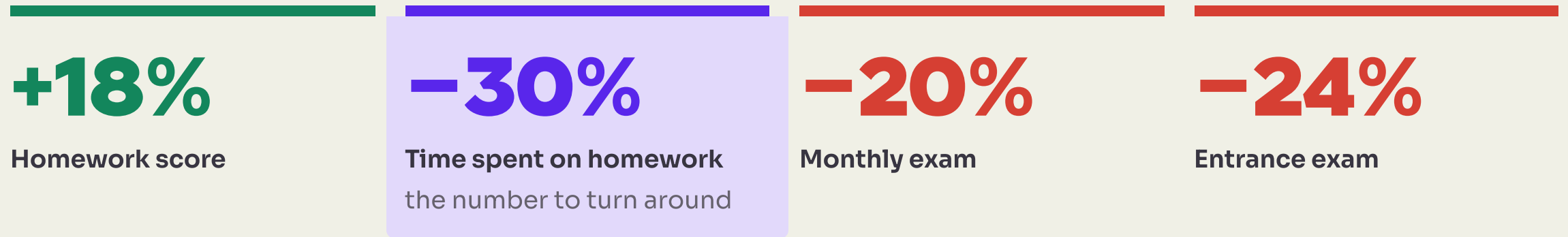
Tamási, *The Power of Outsourcing*, The European Sociologist 55, 2026

“Why would I bother?”

How do I get them to do their own work?

What happens to the time students save?

26,811 Chinese secondary school students, 30 months



Students who kept spending the time barely lost ground. **The lever is time on their own work.**

Two programs, both built with AI

Vibe coding raises the floor, agentic engineering raises the ceiling.

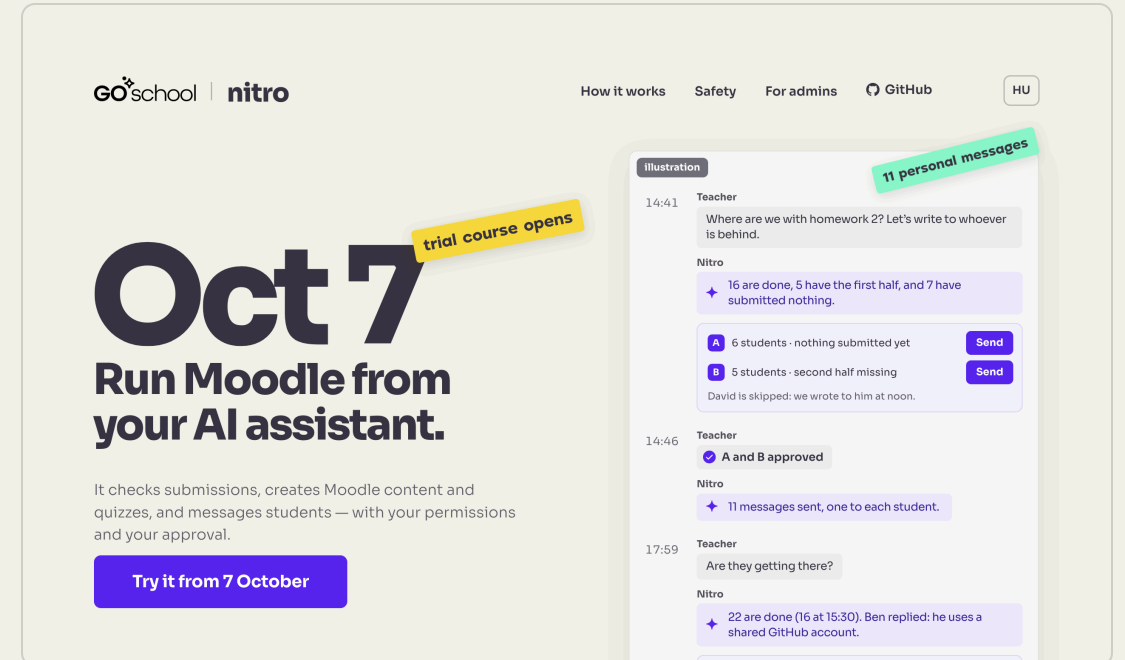
Vibe coding

a playable 3D world from one sentence



Agentic engineering

a Moodle plugin that teachers rely on



Same AI, different way of working

Advanced Programming, 2025/26

Vibe coding

prompt

→

accept

→

it runs

A playable game in minutes.

In my course: the entry task, not the achievement.

Agentic engineering

spec

→

diff

→

test

→

review

Software others can rely on.

The responsibility stays human.

What has to become visible?

Specification

What did the student change in it?

Test

Which result exposed a bug?

Debugging

How did they find and fix it?

Judgement

When did they correct or reject the AI's suggestion?

A reason to bother: their own problem

Advanced Programming, 2025/26 · course design trial, not an impact study

The –30%, turned around.

Goal: more time on their own work, not less. They choose the problem and defend every decision.

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- 01 Time-series store fed by a live air-traffic data stream
 - 02 Difference-in-differences lab on synthetic data
 - 03 Mapping the differences between survey waves
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And the teacher?

**150 students × 4 pieces of evidence
= 600 to read, per assignment.**

It needs support too: time, tools, an institutional framework.

Don't fight offloading. Use the time it frees.

Which of your tasks would make that
time worth the student's own work?



arpadtamasi.github.io/innofair-2026

discussion